

(2) Hick's Law:-

Increase the number of choices will increase the decision time.

This law will be apply if it is sated.

After-Mids:- (CLO-2)

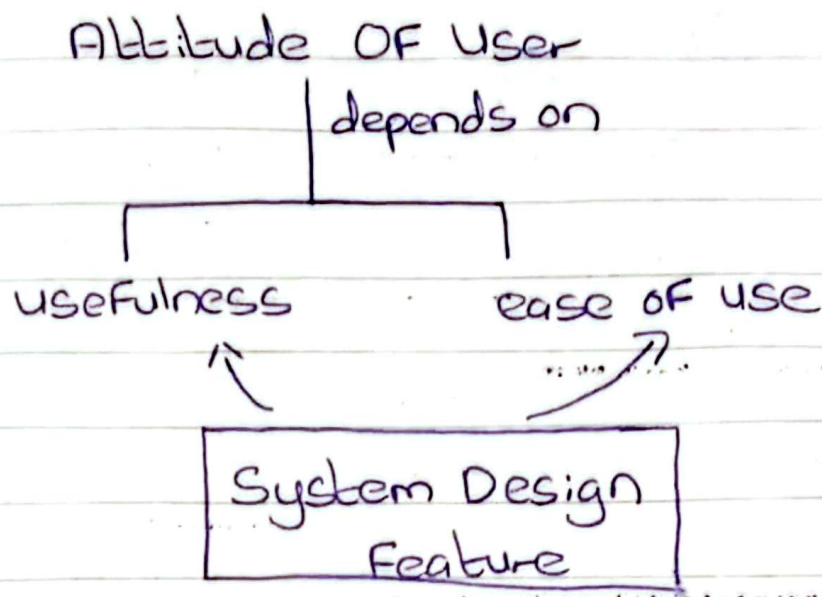
User-Acceptance:-

Perceived useFulness > ease of use

~~FAM~~ model is proposed that

Design Features increase User Acceptance
User Acceptance depends on;

- Attitude of User which depends on perceived (what user feels) useFulness and perceived ease of use. These two things ~~increases~~ impacts the System Design Feature.



- Behaviour is the Subjective likelihood that performing the behaviour will lead to a Specific outcome. The evaluation of Behaviour becomes Attitude.

Usefulness is that the individual believes using a particular System would enhance his or her Job Performance.

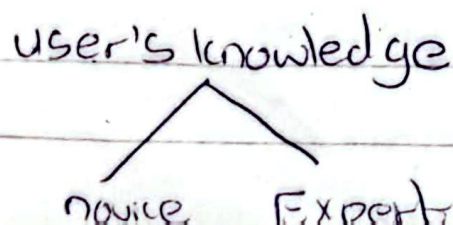
Ease OF use is that an individual believes that using a particular System would be Free of Physical and mental effort.

According to Davis 1989, ease of use and usefulness are distinct features. Ease of use impacts usefulness but vice-versa is not possible. People find easy to use systems more useful. In order to make system useful, you have to add new functional features or make easier for the existing functions. The user can use difficult interface to achieve their tasks but if the easier interface cannot solve or complete task, user will not use it.

Applied Psychology OF user (CLO-1)

The Psychology of the user centers on the Cognitive Abilities. (Learning, Processing, Abilities)

user's behavior = user goals + Task structure +
user's knowledge & Processing
Limits (short term Memory
tendency to make errors)



To understand user's behavior the first step is to understand user's goals.

After that there is a need to know the steps that user's like or not like to perform (Task Structure). One of the most important part of Task structure is UI.

Dimensions OF Behaviour:-

- Functionality:
- Learning OF Tasks
- Time (time taken to perform task)
- Error
- Quality (outcome OF Task's quality)
- Robustness (comfort in performing new & unexpected tasks)
- Acceptability (User Acceptance)

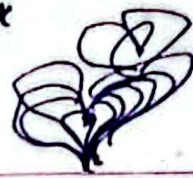
These dimensions depends on type OF users.

Universal Usability:- (CLO-3)

- In a Fair Society, every individual has a right to take the benefits from the Technology. irrespective of Factors like race, religion, age, disability & other Similar Factors.
- ~~The expert user~~ Designing For the expert users is difficult but designing For the large audience that are un-skilled is much more difficult.
- To achieve the highest level of Success, you one have to work on InterFace and information.

There are three challenges to attain universal usability;

- i) Technology Variety
- ii) User Diversity
- iii) Gap in User's knowledge



Participatory Design:- (CLO-1)

It is the design approach that involves the users in the design process.

It is also known as cooperative design, co-design or community design. It is also used in Architecture, Urban planning & Product Design.

Principles:-

- Inclusion
- Empowerment
- collaboration
- Iteration
- Contextual understanding
- user advocacy

It is used in diverse Audience

Inclusion:-

Pick the audience from the diverse audience that may include end-users, designers, developers, domain-experts

collaboration:-

Empowerment:-

Iteration:-

contextual understanding:-

user-advocacy:-

Participatory design is a natural complement to UX design because it actively receives the feedback about the design.

The technique used to implement Participatory design is Contextual Design.

- The CD process stresses the use of customer data to drive the design process.
- This is done in the beginning with collection of data through ethnographic techniques of observation and interviewing. Later, the user is invited to be co-designer by engaging them in dialog over for refinement process using paper prototypes.

Step-1: Contextual Inquiry:-

- Data collection through observation & interviewing
- user will educate the designer about their workflow of the system.

Step-2: Work-Modeling:-

- Format the requirements in any language and visualize the data collected.

Models:-

- (i) Flow-Model:- It tells us about the roles and lines of communication
- (ii) Sequence-Model:- step by step actions a person takes to complete Task.
- (iii) Artifact Model:- Represents the physical or digital objects people use to do work.

Consolidation Step-3:-

- Consolidation includes the Process to Summarize and conclude the different models of user in a single model.

Step-4 visioning: creates a shared picture of ideal Future System based on user needs.

Step-5: User Environment Design:-

organize the System Functions showing how users will interact with system.

Step-6: Paper-Prototyping

uses simple paper sketches to quickly test.

CLO-3:-

- User Analysis
- Task Analysis
- Prototyping
- Evaluation

User-Analysis:-

we can identify user groups by using Matrix Technique. This method is used as;

- Establish Key Audience
- identify what we know & what we lack.
- Identify Features

It is transforming tool to make sense of User Patterns.

- First step is to create 3×3 Matrix
- Second step is to identify important axes.

Task Analysis (CLO-3):-

Task Analysis helps;

- what are the users goals
- what users actually do to achieve goals
- what experience users bring to tasks
- How users previous knowledge & experience influence.
- How they think
- The workflow they follow to perform their tasks.

It's important to perform task Analysis early in your Process. It also helps several other aspects;

- website req. gathering
- Develop your content strategy
- wireframing & Prototyping
- Perform usability Testing

Types of Task Analysis:-

- Cognitive Task Analysis:- is Focused on understanding tasks that require decision making; Problem-Solving, memory
- Hierarchical Task:- divides high level tasks into sub-tasks.

How To Conduct a Task Analysis:- 9 Steps From Slide;

Hierarchical

Task-List Book:-

- 1: Registration
- 2: Dashboard
- 3: Search Filters
- 4: view book
- 5: Add to cart
- 6: Invoice Generate
- 7: Payment Method
- 8: Recommended Books

Decomposition Diagram From Photo;

The next step is ^{that} ~~dialog~~ in which we list the detail steps. Now next step is dialog analysis.

Dialogue Analysis

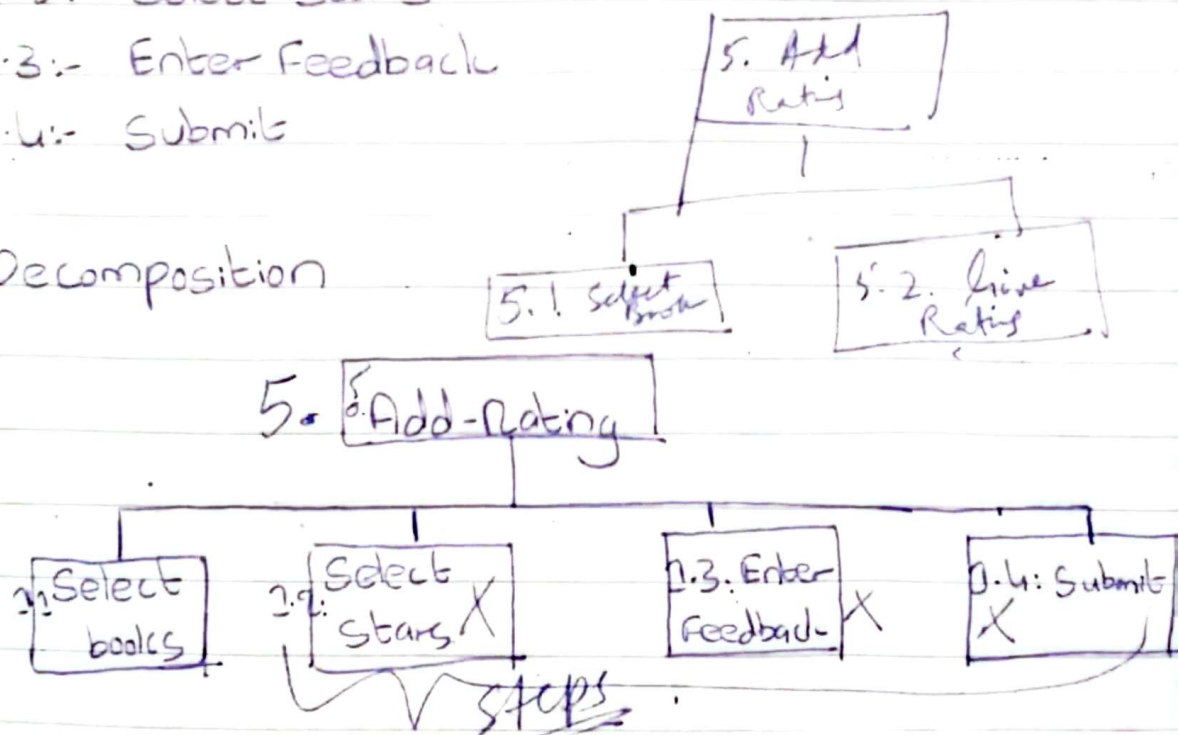
Label	UI-control
- Full Name	Text box
- Address Line 1	Text area
- Address Line 2	"

country	drop-down
city	"
town	"
Post code	text-box
Phone-Number	text-box
Submit	button

1. Add - Rating:-

- 1.1:- Select book
- 1.2:- Select stars
- 1.3:- Enter Feedback
- 1.4:- Submit

Decomposition



Steps of User Interaction:-

1. Move the cursor to the book
2. click the book icon
3. Move cursor to the Feedback portion.

Prototyping (CLO-3):-

It is a draft version of a product that allows you to explore your ideas and show intention behind feature.

- Low Fidelity:- Paper-Prototype
- high Fidelity:- Interactive Prototype

changes done in low Fidelity will have less working as compare to high Fidelity.

wireFrames show Structure of Pages & workflow;

Ethnography

- It is the method for studying users in their respective field.
- The goals are to study details, while the goal of system design are to produce Abstraction.

The Techniques are as follows;

- observing users
- Asking users
- Asking Experts

(i) Observing Users:-

- It involves watching and listening to users.

Scenario (word):-

- The usability consultants observe some problems that are the buttons are small and the screen seems small but the person using is able to use it.
- There are problems in interface but the device is useful.

Answer:-

a) The observer gets the full control in the second scenario where the activity is done in controlled environment.

b) Natural Environment:-

(i) Advantages:-

Here the advantage is that the application is tested independently as the user has the choice to take the steps to perform the activity.

(ii) Disadvantages:-

The disadvantage is that if the small errors or issues are present, the users

Sometimes Ignore them as a result, the observer also slip that ^{without} considering that as Error.

(ii) Control Environment:-

Advantage:-

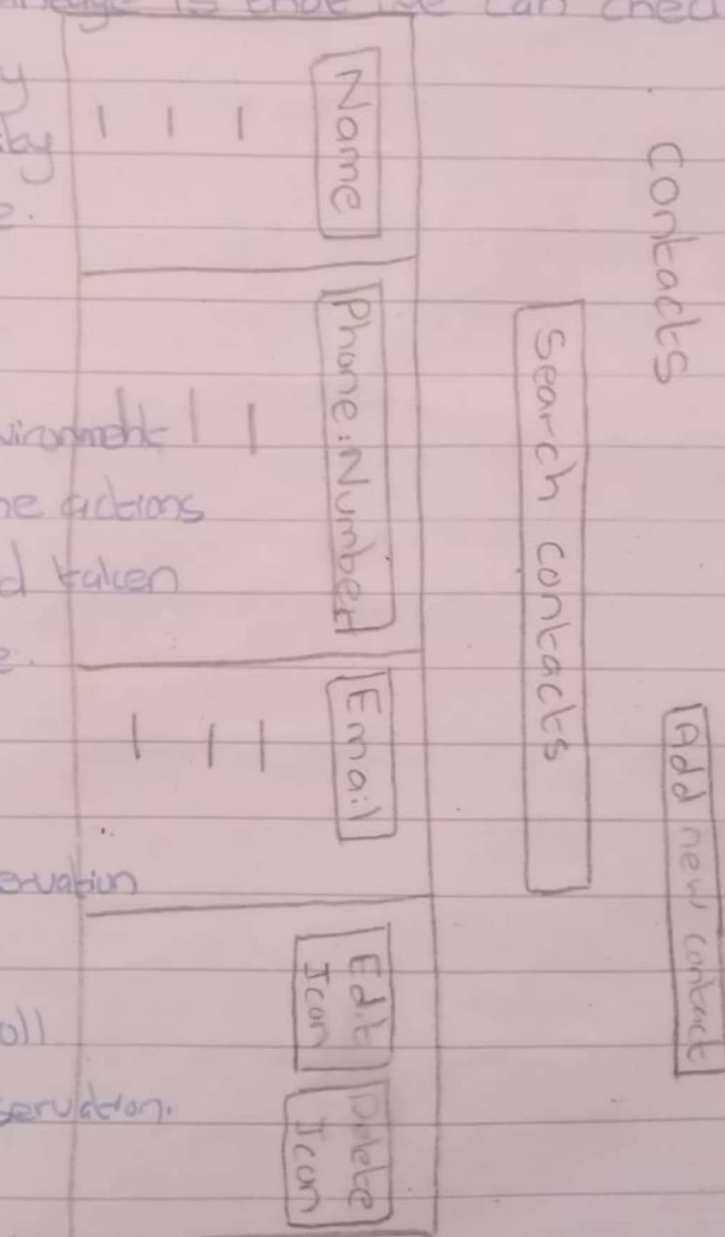
Here the advantage is that we can check the Functionality of each activity had to be done.

Disadvantage:-

Because the environment is controlled, the actions are also limited taken by the Audience.

(i) The natural Environment observation

Should be done after the controll Environment observation.



Thinking aloud Technique is that in which user thinks and talks also.

There are levels of Participants;

- 1 - Complete Participants } users
- 2 - Marginal Participants }

observers who also participate are insider & people who observe from outside are outsider.

Ethnographic slide read after Table given

Evaluation slide revise

Evaluation Technique:-

- Quick & Dirty
- Usability Testing
- Field studies etc
- Predictive Evaluation (Heuristic)

- Quick & Dirty Observation:-

~~- Quick & Dirty~~

- observation in usability testing

- " " Field studies

Fully Participant observe also participant with the user by doing discussion with them.